

Cu-Based Bimetallic Catalysts for Sustainable 1,3-Butadiene and Ethylene Production: A Machine Learning–Accelerated Screening Study

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Abstract: An urgent step toward a sustainable chemical industry is the development of greener, scalable routes to high-value building-block chemicals that avoid the high energy demands of conventional processes. One such target is 1,3-butadiene, an essential precursor for rubbers and plastics, which is currently obtained mainly as a by-product of naphtha steam cracking. This work explores a catalytic alternative: the electrochemical reductive coupling of acetylene using water as the proton source and copper-based bimetallic alloys as catalysts. The catalytic activity of 108 alloy compositions was evaluated using the Nudged Elastic Band (NEB) method within the CatTSunami framework. The same approach was applied to the electrocatalytic synthesis of ethylene, a chemical responsible for 1.2% of global CO₂ emissions, from carbon dioxide. Leveraging Machine Learning Potentials enabled efficient determination of transition-state energies on (111) and (100) alloy surfaces, significantly accelerating the exploration of reaction pathways. Among the alloys screened, Cu₂Pd₂ and Cu₂Rh₂ appear particularly promising for butadiene production, while Cu₂Au₂ emerges among the top candidates for the ethylene reaction. Further mechanistic and experimental validation is required to confirm selectivity and stability under realistic conditions.

Introduction

The electrification of energy-intensive processes that emit greenhouse gases is a pressing challenge that could play a vital role in the decarbonization of the chemical industry.¹ Addressing this challenge requires identifying the most critical processes to target first. Among them, the thermal cracking of petroleum is particularly significant, accounting for about 1.2% of global CO₂ emissions.^{2,3} This process operates at high temperatures and pressures to produce ethylene, with 1,3-butadiene (C₄H₆) generated as a by-product.⁴ Recently, a more cost-efficient process based on shale gas has been developed to produce the highly valuable building-block ethylene.⁵ This new method is still a very energy intensive process, leading to extensive studies on the greenification of this industry sector, leading to the identification of some promising alternative routes.^{6,7} However, a crucial problem remains: since over 95% of the C₄H₆ is obtained from petroleum cracking,⁴ replacing the conventional ethylene production to shale gas will require a scalable synthesis route of C₄H₆. This is especially important because C₄H₆ is a value-added chemical widely used in the industrial production of rubbers and certain plastics.⁸ It has also recently been subject to studies for new industrial applications for its many relevant transformations.⁹

To diversify the chemical industry and reduce society's reliance on oil-derived products, acetylene (C₂H₂) presents itself as a promising alternative with an extremely diverse reactivity.^{9,10} Though its production process is historically very energy intensive, with temperatures exceeding 2'000°C, recent environmental and economic assessments showed that acetylene produced from biochar and decarbonized energy can be sustainable.¹¹ Cheng *et al.* further reported that a life-cycle assessment of an electrocatalytic route to C₄H₆ from C₂H₂ may indeed be both sustainable and low-carbon.¹² These insights highlight acetylene as a viable feedstock for greener chemical processes, provided that selective and scalable catalytic routes can be established.

Until recently, however, no selective synthesis routes were known for C₄H₆ from C₂H₂. Teh *et al.* addressed this gap by developing a catalyst for the electrochemical reduction of C₂H₂ to C₄H₆.¹³ While highly selective, their approach lacked scalability, prompting follow-up work by Cheng *et al.*¹² as well as Bu *et al.*¹⁴ These latter studies proposed catalysts with significantly higher butadiene production rates per hour, albeit at the expense of lower Faradaic efficiency. The aim of the present work was to build on these advances and to design a catalyst for this reaction that would be more cost-effective at the scale required to meet industrial demand for 1,3-butadiene.

To this end, bimetallic alloys were chosen as candi-

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dates. These materials are easily synthesized, can be relatively inexpensive depending on the metals used and, importantly, offer distinct surface reactivities in close proximity, depending on which active site the adsorbates bind to. This property is essential for catalyzing the reaction, as demonstrated by Teh *et al.*¹³ Furthermore, thanks to the mechanism scheme proposed by Cheng *et al.*,¹² a computational screening becomes feasible.

Here, a catalytic screening study using the recently developed CatTSunami framework of the OC20 group to evaluate the catalytic activity of copper-based bimetallic alloys for the 1,3-butadiene synthesis from acetylene and the ethylene synthesis from carbon dioxide was conducted. The CatTSunami framework was used to calculate Nudged Elastic Bands (NEBs) in order to obtain the minimum energy path, the structure, and the energy of the Transition State (TS).¹⁵ The screening covers 208 surfaces constructed from 108 copper-based bimetallic alloys. The (111) surface was chosen for its lower surface energy, stability and particular geometric effects and the (100) surface for its increased chemical reactivity.^{16,17}

It was found for the production of C_4H_6 that Cu_2Pd_2 is a very promising alloy: it shows roughly twice the predicted activity of Cu_4 , and its (111) facet is notable not for a single outstanding barrier but for presenting consistently favorable activation energies across multiple pathways. Cu_2Rh_2 is also worth mentioning for exhibiting the lowest meaningful NEB barrier (0.094 eV) with a highly symmetric transition state. Cu_2Au_2 emerged among the top candidate for the ethylene reaction, however, all candidate alloys require more exhaustive mechanistic sampling and experimental validation (selectivity and operando surface structure) before definitive recommendations can be made.

Results and discussion

Alloy screening: 1,3-butadiene synthesis

Preliminary tests compared Broyden–Fletcher–Goldfarb–Shanno (BFGS) and limited-memory BFGS optimizers, varied spring constants, and evaluated Image-Dependent Pair Potential (IDPP) interpolation versus linear interpolation schemes. The L-BFGS optimizer, IDPP interpolation with short NEB distances and a large spring constant provided the best balance between accuracy, convergence, and computational efficiency, and was therefore adopted for the high-throughput screening.

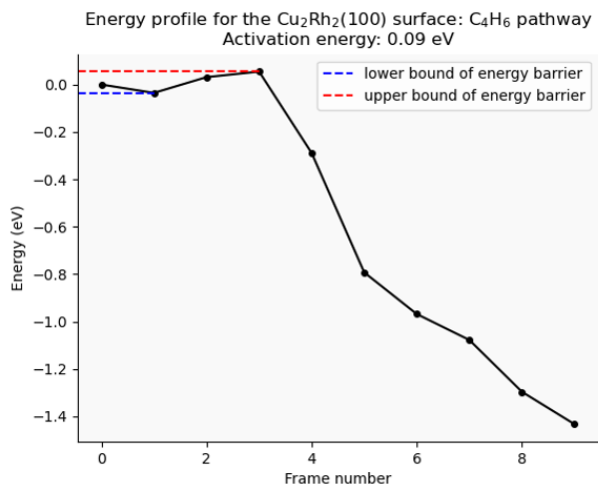
The high-throughput screening of copper-based bimetallic alloys for the electroreduction of acetylene to 1,3-butadiene represents the most computation-

ally intensive and critical component of this study, requiring 25.7 GPU-days of simulation. This exhaustive effort identified the 20 most promising alloys, ranked here in order of decreasing predicted activity: Cu_2Pd_2 , Cu_3Ir , Cu_3Nb , Cu_3Tc , Cu_3Ru , Cu_3Rh , Cu_2Au_2 , Cu_2Zn_2 , Cu_2Pb_2 , Cu_3Os , Cu_2Rh_2 , Ni_2Cu_2 , Cu_2Ga_2 , Cu_3Mn , Cu_2Cd_2 , Cu_3Ni , Cu_3Al , Al_2Cu_2 , Ga_3Cu , and Tl_3Cu . Comprehensive data for each alloy surface on activation energies and estimated rate constants are provided in Table 4 in the Appendices. We observe that the majority of the identified alloys contain 50% or 75% copper, highlighting that copper-rich surfaces achieve the best balance of binding strength and reactivity in accordance with the Sabatier principle.

Relative to previous work by Cheng,¹² the activation energies obtained for pure copper in this study are noticeably lower. This discrepancy may arise from the use of the ML-accelerated workflow, which enables more extensive sampling of candidate pathways but may also introduce systematic differences compared to conventional DFT-only treatments. At the same time, no viable NEB pathway was identified for the $Cu_4(111)$ surface, which is the most stable facet of copper and therefore the most relevant under realistic conditions. This absence may explain the experimentally low catalytic activity of pure copper for C_4H_6 formation, although some viable pathways were found on the less stable $Cu_4(100)$ facet. Whether this reflects a true mechanistic limitation or simply insufficient sampling of initial configurations remains unclear.

Among the candidates, Cu_2Pd_2 stands out as especially promising, with a predicted activity approximately twice that of pure copper. This alloy is particularly attractive because the $Cu_2Pd_2(111)$ surface, which is both the most stable and generally the most representative facet under reaction conditions, also appears highly active in the present analysis.¹⁷ Notably, this surface is effective not because it exhibits exceptionally low activation energies for any single step, but because nearly all reaction pathways have reasonably favorable barriers, collectively contributing to a high overall C_4H_6 production.

By contrast, Cu_2Rh_2 exhibited the lowest meaningful NEB energy barrier of 0.094 eV, with a well-defined TS and a smooth reaction path (Figure 1a), highlighting its favorable kinetics for specific steps even if its overall flux of butadiene is slightly lower than that of Cu_2Pd_2 . The observed TS for Cu_2Rh_2 is particularly meaningful, as it exhibits a near-perfect symmetry of the two $*C_2H_3$ moieties adsorbed on the (100) surface, with the copper atoms forming the top layer and Rh atoms below (Figure 1b,1c). This arrangement suggests a highly balanced interaction



(a) TS of the coupling of two *C_2H_3 molecules on a $Cu_2Rh_2(100)$ surface.

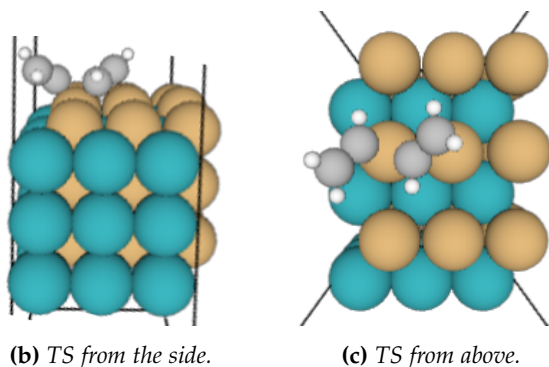


Figure 1: Results of the best NEB path observed for the coupling of two *C_2H_3 molecules on a $Cu_2Rh_2(100)$ surface.

between the reactants and the alloy surface, providing a geometrically favorable pathway for coupling.

Although these results are preliminary, they point to a clear opportunity: alloying can lower the energetic cost of acetylene conversion by tuning intrinsic activity. Recent studies indicate that electrochemical routes to C_4H_6 could already require less energy than conventional cracking, and our high-throughput screening identified Cu-based alloys that could further improve catalytic efficiency.^{12,14} These alloy candidates might be combined in future work with surface-engineering strategies such as wettability control of Cu nanoarrays to optimize both intrinsic activity and surface accessibility.¹² The practicality of the current results must however be interpreted with care: several top candidates contain scarce and costly elements (e.g. Pd, Rh), raising concerns about environmental and economic scalability.

A more rigorous mechanistic treatment will be necessary to fully substantiate these findings. Cheng et al.¹² proposed an expression for estimating the flux of C_4H_6 as a function of the selected pathway, underscoring the importance of resolving the complete

reaction network. In the present work, not all coupling reactions shown in Figure 2 were considered, and several simplifying assumptions were made: the flux of butadiene along each pathway was taken to be proportional only to the exponential term of the Arrhenius equation (Eq. 1), the estimated rate constant was treated as the sum over all pathways (Eq. 2), and the total rate constant was averaged across the (111) and (100) surfaces (Eq. 3). These approximations introduce potential biases. For instance, the initial hydrogenation step may occur nearly instantaneously, reducing the effective contribution of the C_4H_4 pathway. Moreover, because the (111) surface is thermodynamically more stable than the (100) facet, it should in principle carry greater weight in the overall rate constant, but quantifying such a contribution would require additional stability calculations. Additionally, without a systematic mapping of coupling activation energies, it remains unclear whether competing byproduct channels may also be favored. Thus, while the desired pathway appears to proceed with a low activation barrier, the possibility of parallel byproduct formation cannot be excluded. In particular, experimental studies could reveal a tendency for ethylene to form, however C_2H_2 could also polymerize rather than terminate at C_4H_6 , or C_4H_6 could undergo further hydrogenation to butane. Such outcomes would substantially alter the effective selectivity. Given these uncertainties and the intrinsic complexity of the system, even a comprehensive computational mapping of the network may still leave open questions that only experimental validation under realistic conditions can resolve.

Alloy screening: ethylene synthesis

The high-throughput screening of copper-based bimetallic alloys for the electroreduction of carbon dioxide to ethylene required 19.4 GPU-days of simulation. This effort identified only five slightly favored alloys, ranked in order of decreasing predicted activity: Cu_2Au_2 , Cu_2Ag_2 , Ag_3Cu , Cu_3Au , and Cu_3Bi . Comprehensive data for each alloy surface, including activation energies and estimated rate constants, are provided in Table 3 in the appendices. Most of the identified alloys contain gold or silver, highlighting the exceptional catalytic potential of d^{10} metals and suggesting a possible new avenue for optimizing the Sabatier principle in this system.

The results for pure copper are surprising, as the activation energies are extremely low (approximately 0.1 eV for each pathway on $Cu_4(111)$). Unlike the butadiene reaction, the most favorable alloys here only marginally outperform pure copper, which makes interpretation difficult and calls any qualitative trend into question. Experimentally, ethylene selectivity

on a copper surface is low, a behaviour that may arise from rapid conversion to byproducts or from incomplete coverage of the reaction network in this study. Future work should therefore pursue a more complete mechanistic analysis to evaluate whether these surfaces can serve as selective, well-designed catalysts. Useful directions include (i) developing a kinetic model of the system (ii) mapping byproduct formation to identify competing channels, (iii) optimizing activation energies for key intermediates such as $\text{CO}^* \rightarrow \text{CHCO}^*$ (Figure 3), or even (iv) performing a comprehensive reaction-network study on each surface to gain a full mechanistic understanding.

Crucially, experimental confirmation is required to validate catalytic performance under realistic conditions. In particular, (i) stability under oxidizing environments, (ii) absence of unwanted phase segregation or alloy aggregation, and (iii) that the active surface structure (not only bulk composition) maintains the predicted activity. The Miller indices chosen here follow the CatTSunami framework and prior evidence for high C–C coupling activity, but they do not exhaust the chemical or morphological space; future work should therefore explore additional facets and defect geometries (e.g. kink/step sites and (111)/(100) interfaces) to fully map structure–activity relationships.^{15,18–20} Together, targeted experiments and expanded surface modeling will be necessary to establish whether these alloys can become selective, scalable catalysts for acetylene-to-butadiene conversion.

Conclusions

This work used the CatTSunami framework to perform a high-throughput, NEB-based screening of 208 surfaces derived from 108 Cu-based bimetallic alloys in search of efficient electrocatalysts for the conversion of acetylene to 1,3-butadiene as well as for the conversion of carbon dioxide to ethylene. Leveraging ML-accelerated workflows made it possible to explore a much larger materials and pathway space at a fraction of the cost of brute-force DFT (requiring only tens of GPU-days). The screening identified several promising candidates, with Cu_2Pd_2 standing out for its roughly double predicted activity relative to copper, driven by consistently favorable activation barriers across multiple pathways, particularly on the stable (111) surface. Cu_2Rh_2 stands out for exhibiting the lowest meaningful NEB barrier (0.094 eV) on a (100) termination with a highly symmetric, well-defined transition state. For the ethylene synthesis, Cu_2Au_2 and related d^{10} alloys were identified as favored although their apparent advantage is contingent on more exhaustive mechanistic sampling.

Despite these encouraging leads, important caveats temper the conclusions. The present results rely on a limited set of surface terminations ((111) and (100)) and on a restricted sampling of initial states for NEB searches, meaning that missing transition states (namely on stable facets such as $\text{Cu}_4(111)$ for the 1,3-butadiene synthesis) could alter predicted activities. Mechanistic assumptions in the screening, including simplified flux partitioning and incomplete mapping of byproduct formation, further limit the reliability of the results because no study on the selectivity of the alloys were performed. Practical concerns remain as well: several high-ranking alloys contain scarce and expensive elements (e.g., Pd, Au), raising questions about environmental and economic scalability. Additionally, the catalytic behavior of surfaces under operando conditions including oxidizing environments, adsorbate-driven reconstruction, and possible phase segregation or aggregation may differ substantially from idealized slab models. To address these limitations and advance toward viable catalysts, future work should broaden computational searches to additional surface geometries and defects—including vicinal, stepped, and kinked terminations as well as explicit (111)/(100) interfaces while increasing both the number of adsorbate configurations observed and the number of NEB starting guesses to reduce false negatives. It should also involve exhaustive mechanistic mapping of all relevant coupling and hydrogenation steps, with reaction fluxes quantified rather than inferred from single-step metrics. Beyond computation, top candidates need to be validated experimentally with attention to activity, Faradaic efficiency, stability under realistic conditions, and operando structural characterization. Finally, sustainability and scalability must be assessed through life-cycle analyses of acetylene sourcing and alloy element availability, and complementary strategies such as ligand- or support-mediated surface modification and particle-shape control should be explored to optimize facet and edge densities.

In summary, this study demonstrates that ML-guided NEB screening is a practical and powerful approach to discover alloy catalysts for complex C–C coupling chemistry and identifies concrete, experimentally tractable candidates (notably Cu_2Pd_2 and Cu_2Rh_2) that merit follow-up. However, the path from computational prediction to a deployable, sustainable catalyst requires deeper mechanistic mapping, operando experimental validation, and careful consideration of materials availability and scalability.

Methods

Reaction mechanisms

1,3-butadiene synthesis from acetylene The paper of Cheng *et al.*¹² provided mechanistic insights to the reaction, proposing three coupling pathways to butadiene:

- $\text{*C}_2\text{H}_2$ coupling with $\text{*C}_2\text{H}_2$ ($\text{*C}_4\text{H}_4$ pathway)
- $\text{*C}_2\text{H}_3$ coupling with $\text{*C}_2\text{H}_2$ ($\text{*C}_4\text{H}_5$ pathway)
- $\text{*C}_2\text{H}_3$ coupling with $\text{*C}_2\text{H}_3$ ($\text{*C}_4\text{H}_6$ pathway)

The proposed reaction scheme is shown in Figure 2.

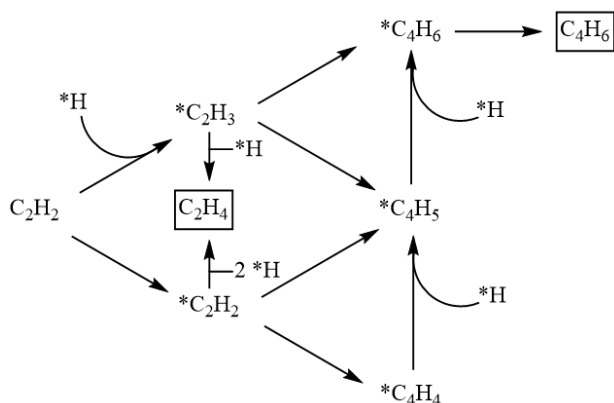


Figure 2: Proposed pathways for the electroreductive coupling of acetylene to 1,3-butadiene and resulting byproducts.

Ethylene synthesis from carbon dioxide Recent studies have proposed a mechanistic framework for the electrochemical synthesis of ethylene from carbon dioxide, summarized in Figure 3. This network encompasses multiple possible C–C coupling steps as well as a competing byproduct pathway. In the present work, we focused on quantifying the activation barriers of the key coupling steps outlined below, as these are thought to represent the rate-determining processes governing ethylene formation.

- *CH coupling with *CH ($\text{*C}_2\text{H}_2$ pathway)
- *CH_2 coupling with *CH ($\text{*C}_2\text{H}_3$ pathway)
- *CH_3 coupling with *CH ($\text{*C}_2\text{H}_4$ pathway)

Pipeline of NEB calculations

The geometries of reactants and products were taken from the adsorbate list proposed by Wander and the DFT study of Cheng *et al.* in the 100% H conditions.^{12,15}

Each reactant and product was independently adsorbed on the surfaces and relaxed using a BFGS optimizer. Forces were provided by a pretrained EquiformerV2 Machine-Learning Potential (MLP)

model (31 million parameters).²¹ The combined initial structure, containing both reactants, was relaxed with the same optimizer and MLP-derived forces. A validity check was done to verify that the atoms were bound as expected. The initial frame was considered only if the inter-reactant distance was lower than the predefined threshold $r_{\text{react,max}}$.

For generating initial NEB paths, candidate initial-final frame pairs were selected based on proximity (within d_{max}) and energy, prioritizing final frames that were both energetically favorable and geometrically close to the initial frame. Linear interpolation was first applied to create a trial path, which was then adjusted to minimize atomic overlap. Paths were penalized for excessive separation while keeping neighboring images evenly spaced. Alternatively, IDPP interpolation was performed using MLP-derived forces. Each interpolation method generated five initial NEB paths made of ten images.

Each NEB calculation used eight intermediate images connected by identical spring constants. Optimization employed a L-BFGS algorithm with force evaluations batched through the MLP model. The Climbing-Image NEB refinement²² was activated once the maximum force perpendicular to the reaction coordinate was below $0.45 \text{ eV/\AA}$. Calculations were considered converged when all forces were below $0.05 \text{ eV/\AA}$. This is consistent with the CatTSunami work.¹⁵

Screening of parameters

To balance accuracy and computational cost, various parameter sets and interpolation schemes were tested on five randomly selected bulk structures across the three reaction pathways. The goal was to determine a spring constant that ensured rapid convergence while maintaining optimal NEB distances, both between initial and final states (d_{max}) and among reactants ($r_{\text{react,max}}$).

Two interpolation methods, linear and IDPP, were evaluated. Only the topmost (111) and (100) surfaces were included during screening. Three spring constants (0.1 , 0.5 , and $1.0 \text{ eV/\AA}^2$) and three initial distance categories (“small,” “medium,” and “large”) were tested. The NEB distances used were different for each reaction, they are presented in Table 1. Intermediate optimization results, including adsorbate relaxations and NEB initial guesses, were stored for efficient parameter screening. A secondary screening on five additional structures was then performed using the optimized parameters and interpolation scheme.

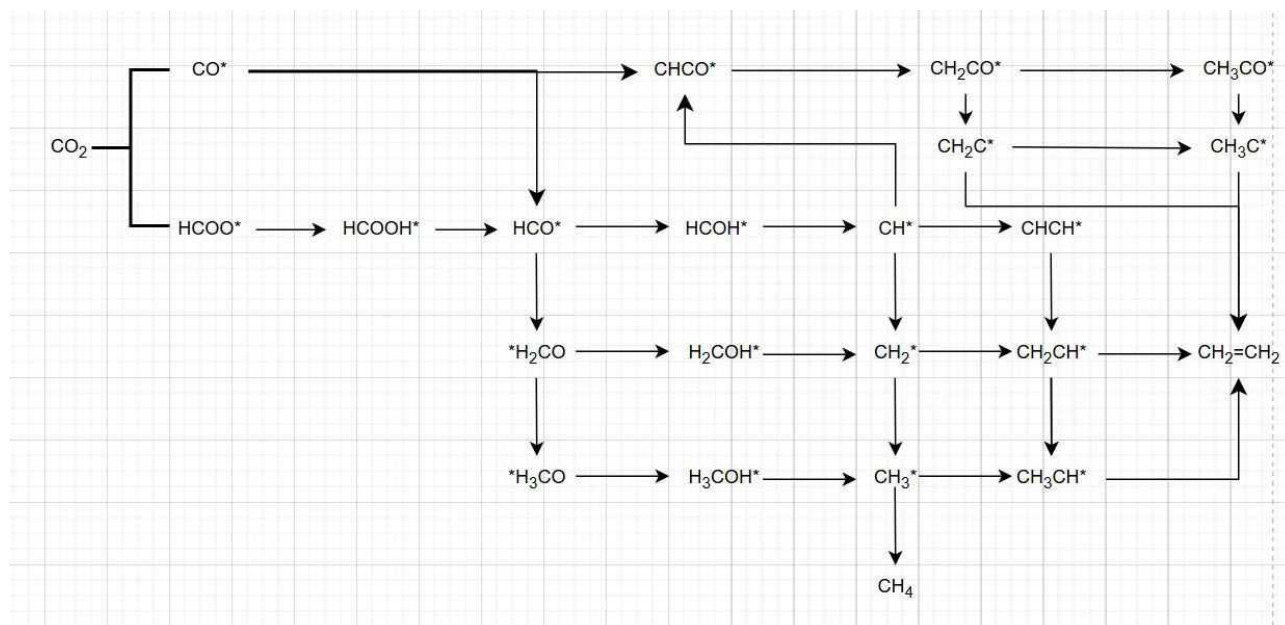


Figure 3: Proposed pathways for the coupling of carbon dioxide to ethylene and resulting byproducts.

Screening of copper bimetallic alloys

A total of 108 copper-based bimetallic alloys were considered. Alloy coordinates were sourced from Mamun *et al.*,²³ comprising of A1 structures (FCC) in proportions of 0%, 25%, 50% or 75%. The bulk structures were cleaved into slabs exposing the (111) and (100) surfaces, with both top and bottom terminations included when distinct.

Each slab comprised six layers, with the bottom three fixed to mimic the bulk and the top three fully relaxed. For the butadiene reaction, all surfaces were evaluated with the NEB pipeline using “small” distances and a “large” spring constant (see parameter screening). For the ethylene reaction from CO₂, the NEB distances are listed in Table 2.

Ranking of copper bimetallic alloys

Alloys were retained only if at least one NEB pathway for the entire alloy exhibited a lower activation barrier than the best copper reference (0.342 eV for butadiene and 0.077 eV for ethylene). The trajectories and energy profiles for each slab and each pathway were manually inspected for physical plausibility, and only the lowest meaningful trajectory was considered. To enable consistent comparison, all coupling steps were treated as contributing equally, with the Arrhenius pre-exponential factor set to unity so that only proportional rate constants were considered. The rate constant for a single pathway i was defined as

$$k_i = e^{-E_{a,i}} \quad (1)$$

where $E_{a,i}$ is the activation energy of pathway i . For a given slab, the overall rate constant was obtained

by summing over all pathways,

$$k_{\text{slab}} = \sum_{i=1}^{N_{\text{pathways}}} k_i = \sum_{i=1}^{N_{\text{pathways}}} e^{-E_{a,i}} \quad (2)$$

Finally, alloys were ranked according to the average rate constant across the (111) and (100) surfaces,

$$k_{\text{bulk}} = \frac{k_{(111)} + k_{(100)}}{2} \quad (3)$$

Based on this metric, the screening identified the 20 top-performing alloys for butadiene and the 5 leading alloys for ethylene.

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Appendices

Table 1: Distances in Angström used in the screening of parameters for the coupling reaction to butadiene.

label	*C ₄ H ₄ pathway		*C ₄ H ₅ pathway		*C ₄ H ₆ pathway	
	$d_{\max}$	$r_{\text{react},\max}$	$d_{\max}$	$r_{\text{react},\max}$	$d_{\max}$	$r_{\text{react},\max}$
"small"	5.0	3.0	6.0	4.0	8.0	6.0
"medium"	6.0	4.0	7.0	4.0	9.0	7.0
"large"	7.0	4.0	8.0	6.0	10.0	8.0

Table 2: Distances in Angström used in the screening of alloys for the coupling reaction to ethylene.

*C ₂ H ₂ pathway		*C ₂ H ₃ pathway		*C ₂ H ₄ pathway	
$d_{\max}$	$r_{\text{react},\max}$	$d_{\max}$	$r_{\text{react},\max}$	$d_{\max}$	$r_{\text{react},\max}$
4.0	2.5	4.0	3.0	4.0	3.0

Table 3: Summary of the most promising copper-based alloys for the catalytic conversion of carbon dioxide to ethylene. Empty activation energy values were set to 100.000 eV. The estimated rate constant k_{slab} was obtained as described in Eq. 2. The k_{bulk} was calculated as in Eq. 3.

Alloy	milller index	$E_a(*\text{C}_2\text{H}_2)$ [eV]	$E_a(*\text{C}_2\text{H}_3)$ [eV]	$E_a(*\text{C}_2\text{H}_4)$ [eV]	k_{slab}	k_{bulk}
Cu ₂ Au ₂	(111)	0.129	0.008	0.173	2.713	2.651
	(100)	0.068	0.213	0.166	2.589	
Cu ₂ Ag ₂	(111)	0.000	0.021	0.292	2.725	2.636
	(100)	0.051	0.287	0.167	2.548	
Ag ₃ Cu	(111)	0.096	0.000	0.193	2.733	2.529
	(100)	0.238	0.262	0.265	2.325	
Cu ₃ Au	(111)	0.087	0.089	0.040	2.792	2.520
	(100)	0.404	0.256	0.216	2.247	
Cu ₃ Bi	(111)	0.107	0.026	0.148	2.734	2.433
	(100)	0.010	0.559	0.564	2.131	
Cu ₄	(111)	0.104	0.077	0.209	2.638	2.319
	(100)	0.312	0.255	0.708	2.000	
Cu ₃ Ag	(111)	0.802	0.199	0.193	2.092	2.284
	(100)	0.245	0.076	0.266	2.476	
Cu ₃ Ga	(111)	0.185	0.068	0.492	2.377	2.250
	(100)	0.278	0.339	0.427	2.122	
Cu ₂ Ga ₂	(111)	0.027	0.441	0.145	2.482	2.049
	(100)	0.032	0.433	100.000	1.617	
Cu ₃ Pb	(111)	0.379	0.375	0.236	2.162	2.010
	(100)	0.211	0.510	0.805	1.857	

Table 4: Summary of the most promising copper-based alloys for the catalytic conversion of acetylene to 1,3-butadiene. Empty activation energy values were set to 100.000 eV. The estimated rate constant k_{slab} was obtained as described in Eq. 2. The k_{bulk} was calculated as in Eq. 3.

Alloy	milller index	$E_a(*C_4H_4)$ [eV]	$E_a(*C_4H_5)$ [eV]	$E_a(*C_4H_6)$ [eV]	k_{slab}	k_{bulk}
Cu2Pd2	(111)	0.205	0.607	0.441	2.002	1.906
	(100)	0.641	0.379	0.516	1.809	
Cu3Ir	(111)	0.811	0.609	0.477	1.609	1.792
	(100)	0.680	0.295	0.324	1.974	
Cu3Nb	(111)	1.133	0.887	0.633	1.265	1.733
	(100)	0.371	0.402	0.172	2.201	
Cu3Tc	(111)	1.238	0.148	0.520	1.747	1.718
	(100)	0.309	0.805	0.678	1.689	
Cu3Ru	(111)	0.684	0.266	0.992	1.642	1.715
	(100)	0.609	0.295	0.695	1.787	
Cu3Rh	(111)	0.834	0.590	0.293	1.735	1.708
	(100)	0.623	0.924	0.291	1.681	
Cu2Au2	(111)	0.390	1.205	0.775	1.437	1.688
	(100)	0.714	0.336	0.309	1.939	
Cu2Zn2	(111)	0.473	1.178	0.961	1.314	1.679
	(100)	0.262	0.523	0.383	2.044	
Cu2Pb2	(111)	0.158	0.496	0.615	2.003	1.674
	(100)	1.208	0.598	0.701	1.345	
Cu3Os	(111)	0.738	0.703	0.277	1.731	1.666
	(100)	0.648	0.250	1.203	1.602	
Cu2Rh2	(111)	0.609	0.922	0.684	1.446	1.639
	(100)	0.639	0.936	0.094	1.831	
Ni2Cu2	(111)	0.695	1.031	1.035	1.211	1.632
	(100)	0.473	0.559	0.152	2.054	
Cu2Ga2	(111)	0.736	1.000	0.898	1.254	1.609
	(100)	0.248	0.182	1.049	1.965	
Cu3Mn	(111)	100.000	0.590	0.520	1.149	1.493
	(100)	0.635	0.773	0.168	1.837	
Cu2Cd2	(111)	0.847	0.945	1.041	1.170	1.463
	(100)	0.486	0.848	0.340	1.755	
Cu3Ni	(111)	0.836	0.637	100.000	0.962	1.459
	(100)	0.605	0.512	0.211	1.955	
Cu3Al	(111)	0.953	1.160	0.801	1.148	1.439
	(100)	1.076	0.504	0.242	1.730	
Al2Cu2	(111)	0.143	0.578	0.949	1.815	1.368
	(100)	1.043	1.016	1.578	0.921	
Ga3Cu	(111)	100.000	100.000	1.365	0.255	1.137
	(100)	0.234	0.447	0.531	2.018	
Tl3Cu	(111)	100.000	100.000	100.000	0.000	0.962
	(100)	0.272	0.373	0.746	1.924	
Cu4	(111)	100.000	100.000	100.000	0.000	0.837
	(100)	0.916	0.572	0.342	1.675	